

MATH 2123 - MATHEMATICS III

2023 Exam Paper: Beginner Guide + Full Solutions

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Designed for a student who is NOT yet an expert.

Each set includes: Topic -> Basics -> Problem-solving procedure -> Full solution -> Exam tips.

How to use this guide

1. Read the Basics before the worked solution.
2. Memorize the Procedure as a reusable recipe, not the numerical answer.
3. Follow each solution one line at a time and check why that step is legal.
4. Before the exam, revise the green Final Answer boxes and the rapid-revision pages.

Paper map - what the 8 sets are really testing

Set	Main topic
1	Vector calculus: gradient and directional derivative, irrotational field/potential, orthogonal surfaces
2	Green theorem/circulation, surface flux on a cylinder, Laplace transforms
3	Divergence volume integral, inverse Laplace, convolution theorem
4	Heaviside/partial fractions, ODE by Laplace transform, line integral
5	Matrix definitions, inverse and elementary matrices, linear dependence
6	Rank, echelon/canonical/normal forms, inverse, linear system with parameter
7	Homogeneous systems, eigenvalues/eigenvectors and eigenvalue-shift property
8	Space curves: T-N-B frame, curvature and acceleration; Newton's law

The 2023 paper mixes vector calculus and Laplace topics in Section A, then matrices/eigensystems and curve mechanics in Section B.

SET 1 - Gradient, Irrotational Flow and Orthogonal Surfaces

What this set tests

- 1(a) Maximum directional derivative and the direction of the gradient.
- 1(b) Curl-free (irrotational) vector fields and scalar velocity potential.
- 1(c) Orthogonal intersection of two surfaces using gradient normal vectors.

Basics you need

- For a scalar field $\phi(x,y,z)$, $\text{grad } \phi = (\phi_x, \phi_y, \phi_z)$.
- The maximum directional derivative at a point is $|\text{grad } \phi|$, and its direction is parallel to $\text{grad } \phi$.
- For $F=(P,Q,R)$, irrotational means $\text{curl } F=0$: $P_y=Q_x$, $P_z=R_x$, $Q_z=R_y$.
- If $\text{curl } F=0$ in a suitable region, $F=\text{grad } \psi$ for a scalar potential ψ .
- A surface $f=0$ has normal $\text{grad } f$. Two surfaces meet orthogonally when $\text{grad } f \cdot \text{grad } g=0$ at their intersection point.

Reusable procedure

Directional derivative: compute grad $\rightarrow$ substitute the point $\rightarrow$ use direction information $\rightarrow$ use its magnitude.

Irrotational field: calculate curl /cross-partial $\rightarrow$ if zero, integrate one component and match the others.

Orthogonal surfaces: use the point-on-surface condition first $\rightarrow$ compute both gradients $\rightarrow$ set their dot product to zero.

1(a) Find a, b, c from the maximum directional derivative

Given $\phi = axy^2 + byz + cz^2x^3$. At $(1,2,-1)$, the maximum directional derivative has magnitude 64 and is parallel to the z-axis.

$$\phi_x = ay^2 + 3cz^2x^2$$

$$\phi_y = 2axy + bz$$

$$\phi_z = by + 2czx^3$$

At $(1,2,-1)$:

$$\text{grad } \phi = (4a+3c, 4a-b, 2b-2c)$$

For this vector to be parallel to the z-axis, its x and y components must vanish:

$$4a+3c=0 \quad \rightarrow \quad c=-4a/3$$

$$4a-b=0 \quad \rightarrow \quad b=4a$$

The remaining z-component becomes:

$$2b-2c = 2(4a+4a/3) = 32a/3$$

$$|\text{grad } \phi| = |32a/3| = 64 \quad \rightarrow \quad |a|=6$$

Under the usual exam convention that the maximum direction is +z, take $32a/3=64$, hence $a=6$.

Final answer

Standard expected answer: $a=6$, $b=24$, $c=-8$. If the word "parallel" allows either +z or -z, the sign-reversed set $a=-6$, $b=-24$, $c=8$ also gives magnitude 64.

Common mistake / exam note

"Maximum directional derivative" normally points along grad ϕ . If the examiner explicitly says +z, use the positive set only.

1(b) Test the fluid motion for irrotationality and find velocity potential

$$\mathbf{v} = (y \sin z - \sin x)\mathbf{i} + (x \sin z + 2yz)\mathbf{j} + (xy \cos z + y^2)\mathbf{k}.$$

Let $P=y \sin z - \sin x$, $Q=x \sin z + 2yz$, $R=xy \cos z + y^2$.

$$P_y = \sin z = Q_x$$

$$P_z = y \cos z = R_x$$

$$Q_z = x \cos z + 2y = R_y$$

All three cross-partial conditions agree, so $\text{curl } \mathbf{v} = 0$ and the motion is irrotational.

Now set $\text{grad } \psi = \mathbf{v}$. Integrate $\psi_x = P$ with respect to x :

$$\psi = xy \sin z + \cos x + f(y, z)$$

Match the y -component:

$$\psi_y = x \sin z + f_y = x \sin z + 2yz \rightarrow f_y = 2yz$$

$$f = y^2 z + g(z)$$

Match the z -component:

$$\psi_z = xy \cos z + y^2 + g'(z) = xy \cos z + y^2 \rightarrow g'(z) = 0$$

Final answer

The motion is irrotational. A velocity potential is $\psi = xy \sin z + \cos x + y^2 z + C$.

1(c) Orthogonal surfaces

Surface 1: $\lambda x^2 - \mu yz = (\lambda + 2)x$.

Surface 2: $4x^2 y + z^3 = 4$. They intersect orthogonally at $(1, -1, 2)$.

First use the fact that the point lies on surface 1:

$$\lambda - \mu(-1)(2) = \lambda + 2 \rightarrow \lambda + 2\mu = \lambda + 2 \rightarrow \mu = 1$$

Write $f = \lambda x^2 - \mu yz - (\lambda + 2)x$ and $g = 4x^2 y + z^3 - 4$.

$$\text{grad } f = (2\lambda x - (\lambda + 2), -\mu z, -\mu y)$$

$$\text{grad } f(1, -1, 2) = (\lambda - 2, -2\mu, \mu)$$

$$\text{grad } g = (8xy, 4x^2, 3z^2) \rightarrow \text{grad } g(1, -1, 2) = (-8, 4, 12)$$

$$\text{grad } f \cdot \text{grad } g = -8(\lambda - 2) - 8\mu + 12\mu = 0$$

$$-8\lambda + 16 + 4\mu = 0$$

With $\mu = 1$: $-8\lambda + 20 = 0 \rightarrow \lambda = 5/2$.

Final answer

$\lambda = 5/2$, $\mu = 1$.

SET 2 - Circulation, Surface Flux and Laplace Transform

What this set tests

- 2(a) Green theorem for circulation around a plane closed curve.
- 2(b) Flux through the curved surface of a quarter-cylinder.
- 2(c) Laplace transform properties: frequency/shift and multiplication by t .

Basics you need

- Green theorem: $\text{integral}_C (P \, dx + Q \, dy) = \text{double integral}_D (Q_x - P_y) \, dA$ for positive (counterclockwise) orientation.
- Surface flux: $\text{integral}_S \mathbf{F} \cdot \mathbf{n} \, dS$. A parametrization gives the vector area element $\mathbf{r}_u \times \mathbf{r}_v \, du \, dv$.
- Frequency-shift: $L\{e^{-at}f(t)\} = F(s+a)$.
- Multiplication by t : $L\{t f(t)\} = -dF/ds$.
- Useful transform: $L\{\sin(at)/t\} = \tan^{-1}(a/s)$.

Reusable procedure

Closed planar circulation $\rightarrow$ identify $P, Q \rightarrow$ Green theorem $\rightarrow$ integrate over the enclosed area.

Cylinder flux $\rightarrow$ parametrize with angle and $z \rightarrow$ choose outward cross product $\rightarrow$ dot with $\mathbf{F} \rightarrow$ integrate.

Laplace property question $\rightarrow$ start from a known transform $\rightarrow$ apply shift/differentiation carefully.

2(a) Circulation around the circle

$A = (y-2x)\mathbf{i} + (3x+2y)\mathbf{j}$. The circle is centered at the origin with radius 2, traversed positively.

$$P = y - 2x, \quad Q = 3x + 2y$$

$$Q_x - P_y = 3 - 1 = 2$$

The region is a disk of radius 2, so $\text{area} = \pi(2)^2 = 4\pi$.

$$\text{Circulation} = \text{double integral}_D 2 \, dA = 2(4\pi) = 8\pi$$

Final answer

Circulation = 8π .

2(b) Flux through $x^2 + y^2 = 16$ in the first octant, $0 \leq z \leq 5$

$\mathbf{F} = z\mathbf{i} + x\mathbf{j} - 3y^2z\mathbf{k}$. The required surface is the curved quarter-cylinder.

Parametrize it by $\mathbf{r}(\theta, z) = (4\cos \theta, 4\sin \theta, z)$, with $0 \leq \theta \leq \pi/2$ and $0 \leq z \leq 5$.

$$\mathbf{r}_\theta = (-4\sin \theta, 4\cos \theta, 0), \quad \mathbf{r}_z = (0, 0, 1)$$

$$\mathbf{r}_\theta \times \mathbf{r}_z = (4\cos \theta, 4\sin \theta, 0) \quad [\text{outward}]$$

On the surface $x = 4\cos \theta$, $y = 4\sin \theta$:

$$\mathbf{F} \cdot (\mathbf{r}_\theta \times \mathbf{r}_z) = 4z \cos \theta + 16 \cos \theta \sin \theta$$

$$\begin{aligned} \text{Flux} &= \text{integral}_0^5 \text{integral}_0^{\pi/2} [4z \cos \theta + 16 \cos \theta \sin \theta] \, d\theta \, dz \\ &= \text{integral}_0^5 (4z + 8) \, dz = [2z^2 + 8z]_0^5 = 50 + 40 = 90 \end{aligned}$$

Final answer

Outward flux through the curved quarter-cylinder = 90.

Common mistake / exam note

The k-component does not contribute here because the outward normal to the curved cylinder is horizontal. If top/bottom caps were included, that would change.

2(c) Laplace transforms

(i) $F(t) = e^{-t} \sin t / t$

Start from $L\{\sin t/t\} = \tan^{-1}(1/s)$. Multiplying by e^{-t} shifts s to $s+1$.

$$L\{e^{-t} \sin t/t\} = \tan^{-1}(1/(s+1))$$

Final answer

$$L\{F(t)\} = \tan^{-1}[1/(s+1)].$$

(ii) $G(t) = t e^{-3t} \sin 2t$

First, $L\{e^{-3t} \sin 2t\} = 2/[(s+3)^2+4]$. Then use $L\{t f(t)\} = -dF/ds$.

$$\begin{aligned} L\{G\} &= -d/ds \left\{ 2/[(s+3)^2+4] \right\} \\ &= 4(s+3) / \{(s+3)^2+4\}^2 \end{aligned}$$

Final answer

$$L\{G(t)\} = 4(s+3)/[(s+3)^2+4]^2.$$

SET 3 - Divergence Integral, Inverse Laplace and Convolution

What this set tests

- 3(a) Compute divergence and integrate it over a rectangular box.
- 3(b) Inverse Laplace using partial fractions and a square-root transform.
- 3(c) Convolution theorem for a product of two Laplace transforms.

Basics you need

- For $F=(P,Q,R)$, $\text{div } F = P_x + Q_y + R_z$.
- Partial fractions split a rational function into standard inverse-Laplace pieces.
- $L^{-1}\{1/\sqrt{s+a}\} = e^{-at}/\sqrt{\pi t}$. Include any constant factor outside the square root.
- Convolution theorem: $L^{-1}\{F(s)G(s)\} = (f*g)(t) = \int_0^t f(\tau)g(t-\tau) d\tau$.

Reusable procedure

Divergence integral: find $\text{div } F$ first $\rightarrow$ write the box limits $\rightarrow$ integrate.

Inverse Laplace: factor denominator $\rightarrow$ partial fractions / standard form $\rightarrow$ invert term by term.

Convolution: split the product into two transforms whose inverses you know $\rightarrow$ write the convolution integral $\rightarrow$ simplify.

3(a) Evaluate triple integral of $\text{div } F$ over the box

$F=(2x^2-3z)i - 2xy j - 4x k$. The box is $3 \leq x \leq 4$, $2 \leq y \leq 3$, $1 \leq z \leq 2$.

$$\begin{aligned}\text{div } F &= d/dx(2x^2-3z) + d/dy(-2xy) + d/dz(-4x) \\ &= 4x - 2x + 0 = 2x\end{aligned}$$

$$\begin{aligned}\text{Integral} &= \int_1^2 \int_2^3 \int_3^4 2x \, dx \, dy \, dz \\ &= [x^2]_3^4 (3-2)(2-1) = (16-9)=7\end{aligned}$$

Final answer

The value of the triple integral is 7.

3(b) Inverse Laplace transforms

(i) $(3s+1)/[(s-1)(s^2+1)]$

Use partial fractions:

$$(3s+1)/[(s-1)(s^2+1)] = A/(s-1) + (Bs+C)/(s^2+1)$$

$$3s+1 = A(s^2+1) + (Bs+C)(s-1)$$

$$A+B=0, \quad C-B=3, \quad A-C=1$$

$$A=2, \quad B=-2, \quad C=1$$

$$F(s) = 2/(s-1) - 2s/(s^2+1) + 1/(s^2+1)$$

Final answer

$$L^{-1}\{F(s)\} = 2e^t - 2\cos t + \sin t.$$

(ii) $1/\sqrt{2s+3}$

$$1/\sqrt{2s+3} = (1/\sqrt{2}) * 1/\sqrt{s+3/2}$$

$$\mathcal{L}^{-1}\{1/\sqrt{s+a}\} = e^{-at}/\sqrt{\pi t}$$

Final answer

$$\mathcal{L}^{-1}\{1/\sqrt{2s+3}\} = e^{-3t/2}/\sqrt{2\pi t}.$$

3(c) Use convolution to invert $1/[s^2(s+1)^2]$

Split the transform as $(1/s^2)*(1/(s+1)^2)$.

$$\mathcal{L}^{-1}\{1/s^2\}=t, \quad \mathcal{L}^{-1}\{1/(s+1)^2\}=t e^{-t}$$

$$f*g = \int_0^t \tau (t-\tau)e^{-(t-\tau)} d\tau$$

Put $u=t-\tau$, so the integral becomes $\int_0^t (t-u)u e^{-u}du$.

$$= t \int_0^t u e^{-u}du - \int_0^t u^2 e^{-u}du$$

$$\int_0^t u e^{-u}du = 1-(t+1)e^{-t}$$

$$\int_0^t u^2 e^{-u}du = 2-(t^2+2t+2)e^{-t}$$

$$\text{Therefore } f*g = t-2+(t+2)e^{-t}$$

Final answer

$$\mathcal{L}^{-1}\{1/[s^2(s+1)^2]\} = t - 2 + (t+2)e^{-t}.$$

SET 4 - Heaviside Expansion, ODE by Laplace and Line Integral

What this set tests

4(a) Inverse Laplace by Heaviside/partial-fraction expansion.

4(b) Solve a second-order initial-value ODE using Laplace transforms.

4(c) Parametrize a path and evaluate a line integral; the printed paper contains an endpoint inconsistency.

Basics you need

- For rational functions, decompose into simple factors, then use standard inverse transforms.
- $L\{y'\} = sY - y(0)$, and $L\{y''\} = s^2Y - sy(0) - y'(0)$.
- A line integral is $\int_C \mathbf{F} \cdot d\mathbf{r}$. The path must agree with the stated endpoints.

Reusable procedure

Inverse rational transform: factor $\rightarrow$ decompose $\rightarrow$ complete squares if needed $\rightarrow$ invert.

ODE: transform every term $\rightarrow$ insert initial data immediately $\rightarrow$ solve for $Y(s)$ $\rightarrow$ decompose $\rightarrow$ invert.

Line integral: parametrize the exact path $\rightarrow$ compute $\mathbf{r}'(t)$ $\rightarrow$ substitute into $\mathbf{F}$ $\rightarrow$ integrate $\mathbf{F} \cdot \mathbf{r}'$.

4(a) Inverse Laplace of $(s-1)/[(s+3)(s^2+2s+2)]$

Write the partial-fraction form:

$$(s-1)/[(s+3)(s^2+2s+2)] = A/(s+3) + (Bs+C)/(s^2+2s+2)$$

Solving for the constants gives:

$$A = -4/5, \quad B = 4/5, \quad C = 1/5$$

$$= -4/[5(s+3)] + (4s+1)/[5(s^2+2s+2)]$$

Complete the square $s^2+2s+2=(s+1)^2+1$, and write $4s+1=4(s+1)-3$.

$$F(s) = -4/[5(s+3)] + (4/5)(s+1)/[(s+1)^2+1] - (3/5)/[(s+1)^2+1]$$

Final answer

$$f(t) = -(4/5)e^{-3t} + (4/5)e^{-t}\cos t - (3/5)e^{-t}\sin t.$$

4(b) Solve $y'' - 7y' + 2y = t+4$, $y(0)=0$, $y'(0)=-1$

Let $L\{y\}=Y(s)$. Transform both sides:

$$(s^2Y - sy(0) - y'(0)) - 7(sY - y(0)) + 2Y = 1/s^2 + 4/s$$

$$(s^2-7s+2)Y + 1 = 1/s^2 + 4/s$$

$$Y = (1/s^2 + 4/s - 1)/(s^2-7s+2)$$

A convenient decomposition is:

$$Y = 1/(2s^2) + 15/(4s) - 3(5s-33)/[4(s^2-7s+2)]$$

Complete the square: $s^2-7s+2=(s-7/2)^2-(\sqrt{41}/2)^2$, and $5s-33=5(s-7/2)-31/2$.

$$\text{Use } L^{-1}\{(s-a)/[(s-a)^2-b^2]\} = e^{at}\cosh(bt)$$

$$\text{and } L^{-1}\{1/[(s-a)^2-b^2]\} = (1/b)e^{at}\sinh(bt).$$

$$y(t) = t/2 + 15/4 - (15/4)e^{7t/2}\cosh(\sqrt{41}t/2)$$

$$+ (93/(4\sqrt{41}))e^{7t/2}\sinh(\sqrt{41}t/2)$$

Final answer

$$y(t) = t/2 + 15/4 - (15/4)e^{7t/2} \cosh(\sqrt{41} t/2) + (93/(4\sqrt{41}))e^{7t/2} \sinh(\sqrt{41} t/2).$$

Common mistake / exam note

A quick check gives $y(0)=0$ and $y'(0)=-1$, so the constants are consistent.

4(c) Line integral - note the printed inconsistency

The paper says "from (0,0,0) to (1,1,1)" but then says the straight line joins (0,0,0) to (1,0,0). These cannot both describe the same path. $F=2x\mathbf{i} - 2y\mathbf{j} + xz\mathbf{k}$.

Interpretation A - use the stated overall endpoints (most natural)

Parameterize the straight line from (0,0,0) to (1,1,1): $\mathbf{r}(t)=(t,t,t)$, $0 \leq t \leq 1$.

Then $\mathbf{r}'(t)=(1,1,1)$ and $F(\mathbf{r}(t))=(2t,-2,t)$.

$F \cdot \mathbf{r}' = 3t-2$, so the integral is $\int_0^1 (3t-2) dt = -1/2$.

Interpretation B - use the printed segment (0,0,0) to (1,0,0)

Parameterize $\mathbf{r}(t)=(t,0,0)$, $0 \leq t \leq 1$.

Then $\mathbf{r}'(t)=(1,0,0)$, $F=(2t,-2,t)$, and $F \cdot \mathbf{r}'=2t$.

The integral is $\int_0^1 2t dt = 1$.

Final answer

Because the question statement is internally inconsistent: answer = $-1/2$ if the intended straight line is (0,0,0) $\rightarrow$ (1,1,1); answer = 1 if the printed segment (0,0,0) $\rightarrow$ (1,0,0) is intended.

SET 5 - Matrix Types, Elementary Matrices and Linear Dependence

What this set tests

- 5(a) Definitions and examples of common matrix types.
- 5(b) Inverse and factorization into elementary matrices.
- 5(c) Linear independence/dependence and expressing one vector as a combination of others.

Basics you need

- Skew-symmetric: $A^T = -A$. Every diagonal entry is 0.
- Diagonal: all off-diagonal entries are 0.
- Orthogonal: $A^T A = I$, so $A^{-1} = A^T$.
- An elementary matrix is obtained by performing one elementary row operation on I .
- Vectors are linearly dependent if a nonzero set of coefficients makes their linear combination zero.

Reusable procedure

Definition question: write the defining equation/properties and give the smallest correct example.

Inverse by elementary operations: reduce A to I ; apply the same operations to $I \rightarrow A^{-1}$.

Linear dependence: put vectors as columns $\rightarrow$ determinant (square case) or solve $c_1v_1 + c_2v_2 + c_3v_3 = 0$.

5(a) Definitions with examples

(i) Skew-symmetric matrix

A square matrix A is skew-symmetric if $A^T = -A$.

Example: $\begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$

Two useful properties: diagonal entries are zero; if the order is odd, $\det A = 0$.

(ii) Diagonal matrix

A square matrix is diagonal if $a_{ij} = 0$ whenever i is not equal to j .

Example: $\begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

Products and powers of diagonal matrices are computed entry-by-entry on the diagonal.

(iii) Orthogonal matrix

A is orthogonal if $A^T A = I$, equivalently $A^{-1} = A^T$.

Example: $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

Its rows and columns form orthonormal sets, and $\det A$ is $+1$ or -1 .

5(b) $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$: inverse and elementary-matrix products

First, $\det A = 4 - 1 = 3$, so the inverse exists.

$$A^{-1} = (1/3) \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

Now reduce A to I using row operations:

$$R1 \leftarrow R1 - R2: \quad \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}$$

$$R2 \leftarrow R2 - R1: \quad \begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix}$$

$$R2 \leftarrow (1/3)R2: \quad \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$R1 \leftarrow R1 + R2: \quad I$$

The corresponding elementary matrices are:

$$E1 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, E2 = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, E3 = \begin{bmatrix} 1 & 0 \\ 0 & 1/3 \end{bmatrix}, E4 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$E4 E3 E2 E1 A = I \rightarrow A^{-1} = E4 E3 E2 E1$$

$$A^{-1} = E1^{-1} E2^{-1} E3^{-1} E4^{-1}$$

A column-operation version is also possible. With

$$F1 = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, F2 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, F3 = \begin{bmatrix} 1 & 0 \\ 0 & 1/3 \end{bmatrix}, F4 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix},$$

$$A F1 F2 F3 F4 = I, \text{ so } A^{-1} = F1 F2 F3 F4.$$

Verification:

$$A A^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} * (1/3) \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = I$$

Final answer

$A^{-1} = (1/3) \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, with the elementary-matrix products shown above.

5(c) Linear dependence of a, b, c

$a = (1, 3, 5)$, $b = (2, 5, 9)$, $c = (-3, 9, 3)$. Put them as columns:

$$M = \begin{bmatrix} 1 & 2 & -3 \\ 3 & 5 & 9 \\ 5 & 9 & 3 \end{bmatrix}$$

$$\det M = 0 \rightarrow \text{the vectors are linearly dependent}$$

Solve $\alpha a + \beta b + \gamma c = 0$. One relation is:

$$-33a + 18b + c = 0$$

$$\text{Therefore } c = 33a - 18b$$

Final answer

The vectors are linearly dependent; $c = 33a - 18b$.

SET 6 - Rank, Canonical/Normal Form, Inverse and a Parameter System

What this set tests

- 6(a) Row reduction to echelon/canonical/normal form, rank, and inverse.
- 6(b) Classify a linear system as unique/infinite/no solution as a parameter changes.

Basics you need

- Rank = number of pivots (nonzero rows) in row-echelon form.
- Row canonical form means reduced row-echelon form (RREF).
- Normal form under row and column operations is $\text{diag}(I_r, 0)$. If rank equals n for an $n \times n$ matrix, normal form is I .
- For a square coefficient matrix: $\det A \neq 0$ means a unique solution for every right-hand side.

Reusable procedure

Rank/inverse: perform elimination; count pivots. For the inverse, augment $[A|I]$ and continue until $[I|A^{-1}]$.
Parameter system: check $\det$ of the coefficient matrix first. If it is nonzero, classification is immediately unique for all parameter values.

6(a) Reduce A and find rank/inverse

$A = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$.

Start elimination:

```
R2 <- R2+R1 -> [0, 3, 1]
R3 <- R3-2R1 -> [0, -5, 3]
R3 <- 3R3+5R2 -> [0, 0, 14]
Row-echelon form =  $\begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 14 \end{bmatrix}$ 
```

There are 3 pivots, so $\text{rank } A = 3$. Continue row reduction and the row canonical form is I_3 . Since $\text{rank} = 3$, the normal form is also I_3 .

To obtain the inverse as a by-product, apply the same operations to $[A|I]$. After the first three operations the right side becomes:

$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 5 & 3 \end{bmatrix}$ with left last row $[0, 0, 14]$

Continue: $R_3 \leftarrow R_3/14$; $R_2 \leftarrow R_2 - R_3$; $R_2 \leftarrow R_2/3$; $R_1 \leftarrow R_1 + R_3$; $R_1 \leftarrow R_1 - 2R_2$.

$A^{-1} = (1/14) \begin{bmatrix} 3 & -1 & 5 \\ 5 & 3 & -1 \\ -1 & 5 & 3 \end{bmatrix}$

Final answer

Row echelon form $\begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 14 \end{bmatrix}$; $\text{rank} = 3$; row canonical form I ; normal form I ; $A^{-1} = (1/14) \begin{bmatrix} 3 & -1 & 5 \\ 5 & 3 & -1 \\ -1 & 5 & 3 \end{bmatrix}$.

6(b) Classify and solve the system

Coefficient matrix $A = \begin{bmatrix} 2 & 3 & 12 \\ 0 & 6 & 1 \\ 1 & 3 & 2 \end{bmatrix}$, and $\text{RHS} = (1, 1, k)^T$.

$\det A = -51 \neq 0$

Therefore the system has a unique solution for EVERY real k . There is no value of k giving infinitely many solutions and no value giving no solution.

For the actual solution, eliminate x. Replace row 1 by row1-2row3:

$$-3y+8z = 1-2k$$

$$6y+z = 1$$

Twice the first equation plus the second eliminates y:

$$17z = 3-4k \rightarrow z=(3-4k)/17$$

$$y=(1-z)/6 = (2k+7)/51$$

$$x=k-3y-2z = (23k-13)/17$$

Final answer

Unique for all k: $x=(23k-13)/17$, $y=(2k+7)/51$, $z=(3-4k)/17$. Infinite-solution values: none. No-solution values: none.

SET 7 - Homogeneous Systems and Eigenvalues/Eigenvectors

What this set tests

- 7(a) Condition for a homogeneous square system to have a nontrivial solution.
- 7(b) Core eigenvalue/eigenvector definitions and the effect of subtracting $4I$.

Basics you need

- A homogeneous system $Ax=0$ always has the trivial solution $x=0$.
- It has a nontrivial solution exactly when $\det A=0$ (for a square matrix).
- Eigenvalue definition: $Av=\lambda v$ for some nonzero vector v .
- Characteristic equation: $\det(A-\lambda I)=0$.
- If $Av=\lambda v$, then $(A-cI)v=(\lambda-c)v$. Eigenvectors stay the same when a scalar multiple of I is subtracted.

Reusable procedure

Homogeneous system: form coefficient matrix $\rightarrow$ determinant $\rightarrow$ set determinant=0 for nontrivial solutions $\rightarrow$ solve with a free parameter.

Eigenproblem: $\det(A-\lambda I)=0 \rightarrow$ each $\lambda \rightarrow$ solve $(A-\lambda I)v=0$.

7(a) Find T for trivial/nontrivial solutions

System: $2x-y+3z=0$; $2x+2y+z=0$; $x-4y+Tz=0$.

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 2 & 2 & 1 \\ 1 & -4 & T \end{bmatrix}$$

$$\det A = 6T - 23$$

Because the system is homogeneous, the trivial solution exists for every T .

A nontrivial solution exists when $\det A=0$:

$$6T - 23 = 0 \rightarrow T = 23/6$$

For $T=23/6$, solve the equations. A convenient one-parameter solution is:

$$(x, y, z) = c(-7, 4, 6)$$

Final answer

Trivial solution: for every T . Nontrivial solutions: only for $T=23/6$, with $(x,y,z)=c(-7,4,6)$, c not equal to 0. For $T \neq 23/6$, the trivial solution is unique.

7(b) Eigensystem of $A=\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ and $A-4I$

Five useful facts about eigensystems

- 1. λ is an eigenvalue if $Av=\lambda v$ for some $v \neq 0$.
- 2. Eigenvalues satisfy $\det(A-\lambda I)=0$.
- 3. The eigenspace for λ is $\text{null}(A-\lambda I)$; its nonzero vectors are eigenvectors.
- 4. Sum of eigenvalues (with algebraic multiplicity) equals $\text{trace}(A)$; product equals $\det(A)$.
- 5. If $B=A-cI$, then every eigenvalue shifts from λ to $\lambda-c$, while the eigenvectors are unchanged.

Now compute the characteristic equation:

$$\det \begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix} = \lambda^2 - \lambda - 1 = 0$$

$$\lambda_1 = (1+\sqrt{5})/2, \quad \lambda_2 = (1-\sqrt{5})/2$$

For either root, an easy eigenvector is $v=(\lambda,1)^T$ because $\lambda^2=\lambda+1$.

$$E_{\lambda_1} = \text{span}\left\{\left(\frac{1+\sqrt{5}}{2}, 1\right)\right\}$$

$$E_{\lambda_2} = \text{span}\left\{\left(\frac{1-\sqrt{5}}{2}, 1\right)\right\}$$

For $A-4I$, subtract 4 from each eigenvalue:

$$\mu_1 = \lambda_1 - 4 = \frac{-7+\sqrt{5}}{2}$$

$$\mu_2 = \lambda_2 - 4 = \frac{-7-\sqrt{5}}{2}$$

The corresponding eigenvectors are exactly the same as for A .

Final answer

A : eigenvalues $(1+\sqrt{5})/2$ and $(1-\sqrt{5})/2$ with eigenvectors proportional to $(\lambda,1)$. $A-4I$: eigenvalues $(-7+\sqrt{5})/2$ and $(-7-\sqrt{5})/2$ with the same eigenvectors.

SET 8 - Space Curves, T-N-B Frame and Particle Motion

What this set tests

- 8(a) Unit tangent, principal normal, binormal, curvature/radius, acceleration components and the osculating plane.
8(b) Use $F=ma$ and integrate acceleration to obtain velocity.

Basics you need

- Velocity $v=r'(t)$, speed= $|r'|$, acceleration $a=r''(t)$.
- Unit tangent $T=r'/|r'|$.
- Principal normal $N=T'/|T'|$, and binormal $B=T \times N$.
- Curvature $\kappa=|T'|/|r'|$; radius of curvature $\rho=1/\kappa$.
- Acceleration decomposition: $a=a_T T+a_N N$, with $a_T=d(\text{speed})/dt$ and $a_N=\text{speed}^2 \kappa$.
- The plane containing T and N is the osculating plane; its normal is B .

Reusable procedure

Curve: differentiate $r \rightarrow$ speed $\rightarrow T \rightarrow$ differentiate $T \rightarrow N \rightarrow$ cross product for $B \rightarrow \kappa$ and $\rho \rightarrow$ acceleration components.

Trihedral plane: evaluate r, T, N, B at the requested t ; choose the plane normal according to which two vectors it contains.

Force: $a=F/m \rightarrow$ integrate each component $\rightarrow$ use initial velocity.

8(a) $r(t)=\cos t \mathbf{i} + \sin t \mathbf{j} + \sqrt{3} \mathbf{k}$

$$r'(t)=(-\sin t, \cos t, 0), \quad |r'|=1$$

$$T=(-\sin t, \cos t, 0)$$

$$T'=(-\cos t, -\sin t, 0), \quad |T'|=1$$

$$N=(-\cos t, -\sin t, 0)$$

$$B=T \times N=(0, 0, 1)=\mathbf{k}$$

$$\kappa=|T'|/|r'|=1, \quad \rho=1/\kappa=1$$

Acceleration is:

$$a=r''=(-\cos t, -\sin t, 0)=N$$

$$a_T=d|v|/dt=0, \quad a_N=|v|^2 \kappa=1$$

Final answer

$T=(-\sin t, \cos t, 0)$, $N=(-\cos t, -\sin t, 0)$, $B=\mathbf{k}$, $\kappa=1$, $\rho=1$, $a=N$, $a_T=0$, $a_N=1$.

Trihedral plane containing T and N at $t=\pi/2$

At $t=\pi/2$: $r=(0, 1, \sqrt{3})$, $T=(-1, 0, 0)$, $N=(0, -1, 0)$, $B=(0, 0, 1)$.

The plane containing T and N is the osculating plane, whose normal is B .

$$B \cdot [(x, y, z) - (0, 1, \sqrt{3})] = 0$$

$$z - \sqrt{3} = 0$$

Final answer

The required osculating (triherdal) plane is $z=\sqrt{3}$.

8(b) Velocity under a time-dependent force

$F = 10t \mathbf{i} + 15t^2 \mathbf{j}$, mass $m=5$, and $\mathbf{v}(0)=5\mathbf{k}$.

$$\mathbf{a} = \mathbf{F}/m = 2t \mathbf{i} + 3t^2 \mathbf{j}$$

Integrate acceleration componentwise:

$$\mathbf{v} = t^2 \mathbf{i} + t^3 \mathbf{j} + \mathbf{C}$$

Use $\mathbf{v}(0)=5\mathbf{k}$, hence $\mathbf{C}=5\mathbf{k}$.

Final answer

$$\mathbf{v}(t) = t^2 \mathbf{i} + t^3 \mathbf{j} + 5\mathbf{k}.$$

Rapid Revision - Procedures to Memorize

Topic	Recipe
Maximum directional derivative	Compute $\text{grad } \phi$. Maximum value is $ \text{grad } \phi $ and its direction is $\text{grad } \phi$.
Irrotational / potential	Check $\text{curl}=0$ using cross-partial, then integrate one component and match the others.
Orthogonal surfaces	Use point-on-surface condition; then $\text{grad } f \cdot \text{grad } g=0$.
Green circulation	$Q_x - P_y$ over the enclosed area for positive orientation.
Cylinder flux	Parametrize with angle and height; use outward $r_\theta \times r_z$.
Laplace shift	e^{-at} shifts $F(s)$ to $F(s+a)$.
t multiplication	$L\{t f(t)\} = -F'(s)$.
Convolution	Product in s-domain $\rightarrow \int_0^t f(\tau)g(t-\tau)d\tau$.
Inverse by rows	$[A I] \rightarrow [I A^{-1}]$.
Rank	Count pivots in echelon form.
Homogeneous nontrivial	For square $Ax=0$, nontrivial iff $\det A=0$.
Eigenproblem	$\det(A - \lambda I)=0$, then $\text{null}(A - \lambda I)$.
T-N-B	$T=r'/ r' $, $N=T'/ T' $, $B=T \times N$, $\kappa= T' / r' $.
Force/velocity	$a=F/m$, integrate, then use initial velocity.

Final answers at a glance

Question	Answer
1(a)	Expected +z set: $a=6, b=24, c=-8$; sign-reversed set also fits magnitude if either parallel direction is allowed.
1(b)	Irrotational; $\psi = xy \sin z + \cos x + y^2 z + C$.
1(c)	$\lambda = 5/2$, $\mu = 1$.
2(a)	8π .
2(b)	90 (outward curved-surface flux).
2(c)	(i) $\tan[1/(s+1)]$ (ii) $4(s+3)/[(s+3)^2+4]^2$.
3(a)	7.
3(b)	(i) $2e^t - 2\cos t + \sin t$ (ii) $e^{-3t/2}/\sqrt{2\pi t}$.
3(c)	$t - 2 + (t+2)e^{-t}$.
4(a)	$-(4/5)e^{-3t} + (4/5)e^{-t}\cos t - (3/5)e^{-t}\sin t$.
4(b)	$t/2 + 15/4 - (15/4)e^{7t/2}\cosh(\sqrt{41}t/2) + (93/(4\sqrt{41}))e^{7t/2}\sinh(\sqrt{41}t/2)$.
4(c)	Printed typo: $-1/2$ for straight line to $(1,1,1)$; 1 for

	printed segment to (1,0,0).
5(b)	$A^{-1} = (1/3)[[2,-1],[-1,2]]$.
5(c)	Dependent; $c = 33a - 18b$.
6(a)	rank=3; canonical/normal=I; $A^{-1} = (1/14)[[3,-1,5], [5,3,-1],[-1,5,3]]$.
6(b)	Unique for all k: $x = (23k-13)/17$, $y = (2k+7)/51$, $z = (3-4k)/17$.
7(a)	Nontrivial only $T = 23/6$; solution $c(-7,4,6)$.
7(b)	A eigenvalues $(1 \pm \sqrt{5})/2$; A-4I eigenvalues $(-7 \pm \sqrt{5})/2$; same eigenvectors.
8(a)	$T = (-\sin, \cos, 0)$, $N = (-\cos, -\sin, 0)$, $B = k$, $\kappa = \rho = 1$, $a_T = 0, a_N = 1$; plane $z = \sqrt{3}$.
8(b)	$\mathbf{v} = t^2 \mathbf{i} + t^3 \mathbf{j} + 5\mathbf{k}$.

Exam Strategy for This 2023 Paper

If you are starting from weak basics, prioritize these first

1. Sets 5-7: matrices, row operations and eigenvalues are strongly procedural.
2. Sets 2-4 Laplace parts: learn a short transform table plus shift, t-multiplication, partial fractions and convolution.
3. Set 1: master gradient, curl and orthogonal-surface dot products.
4. Set 8: memorize the T-N-B formulas and practice one circle/helix example.
5. Vector integrals: always draw or describe the path/surface before choosing limits.

In the exam, write the governing formula before substitution. For vector integrals, clearly state orientation and limits. For matrices, show row operations. For Laplace ODEs, insert the initial values immediately after transforming.

Final checking habit

Check signs, endpoints/orientation, transform shifts, row-operation arithmetic, and whether every subpart was answered. The 2023 Q4(c) itself has an endpoint/path inconsistency, so explicitly state the interpretation you use.